

Introduction & Theoretical Framework

Research question, objectives, and the three literature pillars that motivate variable selection

Research Question

Can macroeconomic variables — especially CPI — improve forecasts of Ireland's 10Y government bond yield beyond a univariate ARIMA baseline?

Three Objectives

01

Build 9-variable macro-financial dataset (FRED + Yahoo Finance, 2003–2024)

02

Estimate ARIMAX & SARIMAX — test if CPI and seasonality improve forecasts

03

Use VAR to decompose yield variance — quantify global vs. domestic drivers

Fisher Effect

Mishkin (1992)

Nominal rates move one-for-one with inflation in the long run. Short-run effect absent — yields react to permanent shifts.

Implication:

Motivates CPI as primary exogenous variable; justifies $d=1$ differencing

Fiscal & Global Determinants

Baldacci & Kumar (2010)

Deficits and debt/GDP raise yields. VIX and US 10Y significant even after domestic controls — global factors dominate small open economies.

Implication:

Justifies variable set: US 10Y, VIX, ECB rate, CPI

Long-Run vs Short-Run Dynamics

Poghosyan (2014)

Global factors dominate short run; domestic macro governs long-run equilibrium. Adjustment speed ~30% per year.

Implication:

Informs Cholesky ordering: US10Y → VIX → ECB → CPI → IP → $\tilde{y}_t$

Literature Review & Variable Selection

7 papers across 3 themes | 9 variables | FRED + Yahoo Finance | Monthly, Jan 2003 – Jan 2024

Literature Review — 7 Papers, 3 Themes			
Paper	Focus	Key finding	Contribution here
Mishkin (1992)	Fisher effect — US rates	Long-run Fisher confirmed; short-run absent	CPI as primary exog; I(1) differencing
Baldacci & Kumar (2010)	Fiscal determinants, 31 countries	Deficits, VIX, US yield → higher yields	Core variable set; Cholesky ordering
Poghosyan (2014)	Long/short-run dynamics, 22 AEs	Global factors dominate short run	Lag structure; global vs domestic split
Huei & Kamisan (2025)	CPI forecasting, Malaysia	ARIMAX best in-sample; ARIMA wins OOS	Warns against in-sample-only evaluation
Shinkarenko et al. (2021)	CPI seasonality, Ukraine	Strong seasonality in CPI; weak in finance	ACF/STL methodology; seasonality prior
Are, Aako & Ojo (2023)	Inflation forecasting, Nigeria	SARIMAX best; only interest rate significant	Validates SARIMAX; warns insignif. vars
Zhou (2025)	ARIMA for 10Y yields, US & Singapore	ARIMA(1,1,1); MAPE=1.2% for US yield	Closest comparator — this study extends

9 Variables — FRED & Yahoo Finance					
Variable	Ticker		Source	Sign	Theoretical channel
Ireland 10Y Yield (target)	IRLTLT01EZM156N	—	US 10Y Treasury	DGS10	+
ECB Policy Rate	ECBDFR	+	EUR/USD	EURUSD=X	+/-
Euro Area CPI	CP0000EZ19M086NEST	+	Euro Stoxx 50	^STOXX50E	-
EA Manufacturing IP	EA19PRMNT001GPSAM	+	Brent Crude Oil	BZ=F	+
VIX	VIXCLS	-			

Data Overview & Stationarity Tests

191 clean observations | ADF: H_0 = unit root | $p < 0.05 \rightarrow$ stationary | $I(1)$ series differenced before VAR

Descriptive Statistics				
Variable	Mean	Std	Min	Max
Bond Yield (%)	2.65	1.49	-0.09	4.82
ECB Rate (%)	0.62	1.22	-0.50	4.00
CPI (index)	98.00	10.57	80.14	124.51
VIX	19.25	8.24	10.13	62.67
US 10Y (%)	2.97	1.12	0.62	5.11
EUR/USD	1.24	0.13	0.98	1.58
Stoxx50	3,291	604	1,994	4,509
Brent (USD)	78.61	24.88	26.63	134.10

Stationarity Tests (ADF)		
Bond Yield	p=0.516	$d=1$ in ARIMA(1,1,1)
ECB Rate	p=0.312	First-differenced in VAR
US 10Y	p=0.455	First-differenced in VAR
log EUR/USD	p=0.311	Levels in ARIMAX exog
log Stoxx50	p=0.225	Levels in ARIMAX exog
CPI MoM %	p=0.016	Stationary via <code>pct_change()</code>
IP MoM %	p=0.000	Stationary via <code>pct_change()</code>
VIX	p=0.001	Mean-reverts naturally
log Brent	p=0.031	Borderline stationary

Seasonality Analysis & Model Specifications

STL decomposition · Ljung-Box test · ARIMAX and SARIMAX equations

Seasonality Analysis

STL Seasonal Strength

0.254

MODERATE (0.2–0.5)

Ljung-Box Lag 12

Q=2,581

Autocorrelation present — but from persistence

Ljung-Box Lag 24

Q=4,269

Confirms long-memory in yield levels

ACF spike at lag 12

No clear spike

Smooth decay — no calendar pattern

SARIMAX seasonal terms

All p>0.10

SAR, SMA, Fourier — all insignificant

Verdict: Autocorrelation reflects yield PERSISTENCE, not calendar seasonality. ARIMAX preferred over SARIMAX.

Model Specifications

ARIMAX(1,1,1)

Autoregressive Integrated Moving Average with eXogenous variables

$$\Delta y_t = \phi_1 \Delta y_{t-1} + \varepsilon_t - \theta_1 \varepsilon_{t-1} + \beta' x_t$$

p=1

1 AR lag

d=1

Differences yield (removes unit root)

q=1

1 MA term

$\beta' x_t$

8 macro exogenous predictors

SARIMAX(1,1,1)(1,0,1,12)

ARIMAX + SAR(1)+SMA(1) at period 12, plus Fourier seasonal encoding

$$\text{ARIMA} + \phi_1 y_{t-12} + \theta_1 \varepsilon_{t-12} + \gamma \sin(2\pi m/12) + \gamma \cos(2\pi m/12)$$

P=1,Q=1

SAR+SMA at lag 12

D=0

No seasonal differencing
sin/cos pairs k=1,2 (4 params)

s=12

Annual period

Fourier

ARIMAX(1,1,1) — Full Results

167 training obs (Jul 2007–Jan 2022) | AIC=-134.53 | BIC=-100.30 | Ljung-Box p=0.66 ✓ white noise residuals

Exogenous coefficient estimates with 95% CI

us10y		+0.651	p=0.000 ✓
log_stoxx50		-1.862	p=0.000 ✓
log_eurusd		+1.350	p=0.019 ✓
vix		-0.006	p=0.052
ecb_rate		+0.018	p=0.869
d_cpi		-0.015	p=0.371
log_brent		-0.039	p=0.845
d_ind_prod		+0.000	p=0.460

Model fit

AIC:	-134.53
BIC:	-100.30
Log-Likelihood:	78.27
Ljung-Box p:	0.66 ✓ white noise

US 10Y dominates

$\beta=+0.65$ ($p<0.001$): 1pp rise in US rates $\rightarrow$ 0.65pp rise in Irish yield. Global rate transmission confirmed.

Equity risk channel

$\beta=-1.86$ ($p<0.001$): Stoxx50 rise $\rightarrow$ yield falls. Investors move from bonds to equities when risk appetite rises.

CPI insignificant

$\beta=-0.015$ ($p=0.37$): CPI absorbed by global rate pricing at monthly frequency. Fisher effect operates at longer horizons.

SARIMAX Results & ARIMAX vs SARIMAX Comparison

SARIMAX adds 6 seasonal parameters — none significant | ARIMAX preferred on all key metrics

SARIMAX Seasonal Coefficients — All Insignificant			
ar.L1 (ARMA)	3.72e-05	p=0.999	Essentially zero — optimiser failed
ma.L1 (ARMA)	3.72e-05	p=0.999	Matches ar.L1 — multicollinearity
ar.S.L12 (SAR)	-0.227	p=0.524	Annual AR — not significant
ma.S.L12 (SMA)	-0.092	p=0.807	Annual MA — not significant
sin_1/cos_1	≈+0.04/-0.02	p=0.19/ 0.32	Fourier k=1 — not significant
sin_2/cos_2	≈-0.00/+0.03	p=0.95/ 0.12	Fourier k=2 — not significant

⚠ Covariance matrix near-singular (condition number = 4.51×10^{17}) — standard errors unreliable. Model is over-parameterised for 167 observations.

ARIMAX vs SARIMAX — Head-to-Head				
Metric	ARIMAX	SARIMAX	Winner	Why
AIC (lower = better)	-134.53	-138.47	SARIMAX	In-sample fit slightly better
BIC (penalises parameters)	-100.30	-85.57	ARIMAX	ARIMAX wins on parsimony
Out-of-sample RMSE (pp)	1.031	1.112	ARIMAX	ARIMAX forecasts more accurately
Out-of-sample MAE (pp)	0.956	1.041	ARIMAX	Consistent with RMSE result
Seasonal terms significant?	N/A	All p>0.10	ARIMAX	No calendar effect in yields
Covariance matrix stable?	Yes	No — singular	ARIMAX	SE unreliable in SARIMAX

Verdict: ARIMAX preferred — lower BIC, better OOS RMSE, stable covariance matrix, no over-parameterisation. Seasonality adds complexity without forecasting benefit.

Out-of-Sample Forecast Performance

24-month hold-out: Feb 2022 – Jan 2024 | Compared against random walk (naïve) benchmark

2.34

Naïve RW
RMSE (pp)

1.03

ARIMAX
RMSE (pp)

1.11

SARIMAX
RMSE (pp)

56%

RMSE cut
vs naïve

Both models beat the random walk

Random walk RMSE = 2.34 pp. ARIMAX cuts this to 1.03 pp — a 56% improvement. Macroeconomic variables add genuine predictive power beyond naïve extrapolation.

ARIMAX outperforms SARIMAX out-of-sample

RMSE: 1.031 vs 1.112 pp. MAE: 0.956 vs 1.041 pp. Seasonal component slightly worsens OOS accuracy — consistent with Huei & Kamisan (2025) overfitting warning.

Accuracy in context

MAE of 0.96 pp means the model is off by ~1 percentage point on average over a 2-year horizon. For macro forecasting at monthly frequency, this is a strong result.

Literature comparison

Zhou (2025) reports MAPE of 1.2% for US yields using univariate ARIMA. Our ARIMAX on Irish yields achieves comparable accuracy while incorporating 8 macro predictors.

VAR(2) Analysis — Granger Causality

System of 6 first-differenced stationary series | 238 obs | Cholesky: US10Y→VIX→ECB→CPI→IP→Yield

What is Granger causality?

Variable X Granger-causes Y if past values of X improve predictions of future Y, beyond what Y's own history already provides. Tests predictive power — not economic causation.

Granger causality tests → Ireland 10Y bond yield

CPI (d_cpi)	p=0.020	YES ✓	ONLY significant predictor — CPI changes lead yield changes
US 10Y	p=0.547	No	Moves contemporaneously with yield — doesn't lead it
VIX	p=0.722	No	No predictive power for future yield changes
ECB Rate	p=0.205	No	ECB rate changes don't predict future movements
Ind Prod (MoM)	p=0.635	No	Industrial production has no predictive power

VAR(2) diagnostics

Durbin-Watson (all):	1.95–2.03 ✓ OK
AIC:	1.422
Equations:	6
Observations:	238

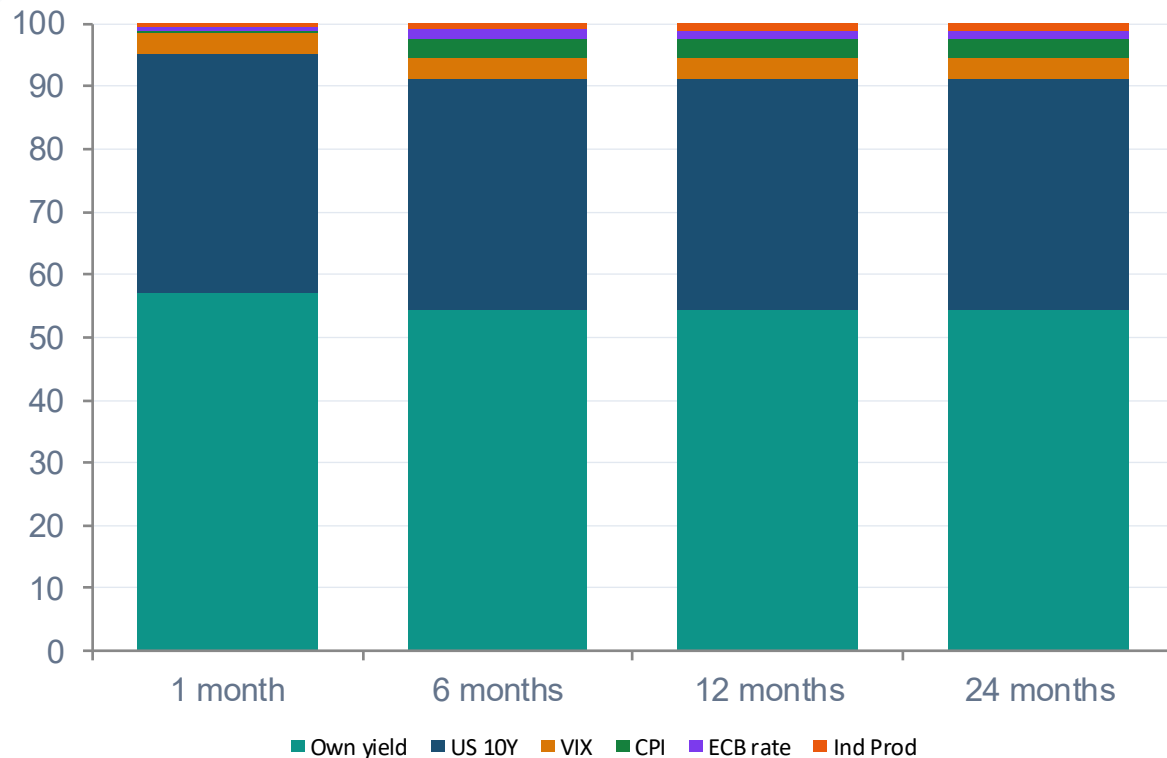
The levels vs. dynamics puzzle

US 10Y is highly significant in ARIMAX (levels, $p < 0.001$) but fails Granger causality in the VAR (first differences, $p = 0.55$). US yields move simultaneously with Irish yields but don't predict future changes.

CPI does the opposite — insignificant in ARIMAX levels ($p = 0.37$) but the only Granger-causal predictor ($p = 0.02$). The Fisher effect operates through dynamics, not contemporaneous levels.

Forecast Error Variance Decomposition (FEVD)

% of Ireland 10Y bond yield forecast error variance explained by each shock | VAR(2), Cholesky identification



At 12-month horizon

Own past yield 54.4%

US 10Y shocks 36.8%

VIX shocks 3.3%

CPI shocks 3.2%

ECB rate 1.3%

Ind Prod 1.0%

US 10Y + own dynamics = 91.2%

Global factor dominance confirmed

(Poghosyan 2014; Baldacci & Kumar 2010)

Research Gaps Addressed & Study Contributions

Three literature gaps — three contributions — three empirical findings

Gap 1	Literature gap No ARIMAX/SARIMAX study uses CPI as an exogenous variable to forecast sovereign bond yields	Our contribution Bridges Huei & Kamisan (2025) (ARIMAX for CPI) and Zhou (2025) (ARIMA for yields) — placing CPI directly inside the bond yield model	Empirical finding CPI insignificant in levels ($p=0.37$) but Granger-causes yield in first differences ($p=0.02$) — Fisher effect operates through dynamics, not contemporaneous levels
Gap 2	Literature gap Small open eurozone economies underrepresented — prior work focuses on US, Singapore, or EM	Our contribution First ARIMAX/SARIMAX application to Ireland's 10Y yield — captures the tension between ECB domestic policy and US global rate dominance	Empirical finding Global factor dominance confirmed: US 10Y + own yield dynamics explain 91.2% of yield variance. ECB rate subsumed by US 10Y in both ARIMAX and FEVD
Gap 3	Literature gap No comparative evaluation of seasonal encoding methods for sovereign yield forecasting	Our contribution Tests month dummies, Fourier terms, and seasonal ARMA simultaneously — enabling direct comparison of all three seasonal specifications	Empirical finding All seasonal specifications insignificant ($p>0.10$). STL strength = 0.254 reflects persistence not calendar effects. ARIMAX preferred over SARIMAX